B. Tech. (Computer Engineering) Vth Semester
2nd Sessional 2024-25, AUTOMATA THEORY (CEN-502)

Time: 1 Hour **MM:15**

Note: Attempt all questions. **Date: 23/10/24**

Your Roll No : .....

Q No.	CO Mapped
1	3
2	3
3	3

Q. No	Question	Marks
1.	Describe Chomsky Classification of formal Languages using suitable diagram. Also explain the concept of Ambiguity in Grammars. Give example of an Ambiguous grammar along with the proof of its ambiguity.	3+2+1
2.	Construct the CFG for the following language with suitable verifications : a. L = { wcw^R : w contains {a,b}*}. [*Where* w^R *stands for reverse of string w.*] b. L2 = { Accepting set of all strings over {0, 1} which do not contain the substring 010} ***You Must Verify your answer with suitable examples.***	4
3	Convert the following CFG to CNF along with mentioning appropriate rules wherever needed. X → ggPH P → gKH | HP | ε H → abH | KK | ε T → t	5

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